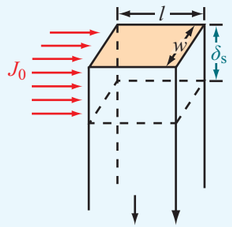


(a) Exponentially decaying $\tilde{J}_x(z)$



(b) Equivalent J_0 over skin depth δ_s

Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

The EM fields at any depth z in the conductor are given by

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} E_0 e^{-\alpha z} e^{-j\beta z}, \tag{7.83a}$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z}. \tag{7.83b}$$

From $\mathbf{J} = \sigma\mathbf{E}$, the current flows in the x direction, and its density is

$$\tilde{\mathbf{J}}(z) = \hat{\mathbf{x}}\, \tilde{J}_x(z), \tag{7.84}$$

with

$$\tilde{J}_x(z) = \sigma E_0 e^{-\alpha z} e^{-j\beta z} = J_0 e^{-\alpha z} e^{-j\beta z}, \tag{7.85}$$

where $J_0 = \sigma E_0$ is the amplitude of the current density at the surface. In terms of the skin depth $\delta_s = 1/\alpha$ defined by Eq. (7.72) and using the fact that in a good conductor $\alpha = \beta$ as expressed by Eq. (7.77b), Eq. (7.85) can be written as

$$\tilde{J}_x(z) = J_0 e^{-(1+j)z/\delta_s} \qquad (\text{A/m}^2). \tag{7.86}$$

The current flowing through a rectangular strip of width w along the y direction and extending between zero and ∞ in the z direction is

$$\tilde{I} = w \int_0^\infty \tilde{J}_x(z)\, dz = w \int_0^\infty J_0 e^{-(1+j)z/\delta_s}\, dz = \frac{J_0 w \delta_s}{(1+j)} \quad (\text{A}). \tag{7.87}$$

The numerator of Eq. (7.87) is reminiscent of a uniform current density J_0 flowing through a thin surface of width w and depth δ_s. Because $\tilde{J}_x(z)$ decreases exponentially with depth z, a conductor of finite thickness d can be considered electrically equivalent to one of infinite depth as long as d exceeds a few skin depths. Indeed, if $d = 3\delta_s$ [instead of ∞ in the integral of Eq. (7.87)], the error incurred in using the result on the right-hand side of Eq. (7.87) is less than 5%; if $d = 5\delta_s$, the error is less than 1%.

The voltage across a length l at the surface (**Fig. 7-15(b)**) is given by

$$\tilde{V} = E_0 l = \frac{J_0}{\sigma} l. \tag{7.88}$$

Hence, the impedance of a slab of width w, length l, and depth $d = \infty$ (or, in practice, $d > 5\delta_s$) is

$$Z = \frac{\tilde{V}}{\tilde{I}} = \frac{1+j}{\sigma\delta_s} \frac{l}{w} \qquad (\Omega). \tag{7.89}$$

It is customary to represent Z as

$$Z = Z_s \frac{l}{w}, \tag{7.90}$$

where Z_s is called the ***internal*** or ***surface impedance*** of the conductor and is defined as the impedance Z for a length $l = 1$ m and a width $w = 1$ m. Thus,

$$Z_s = \frac{1+j}{\sigma\delta_s} \qquad (\Omega). \tag{7.91}$$

Since the reactive part of Z_s is positive, Z_s can be defined as

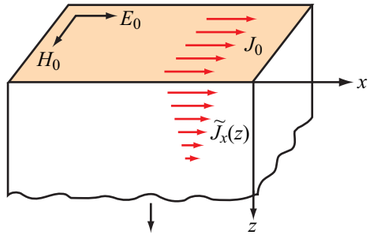
$$Z_s = R_s + j\omega L_s$$

with

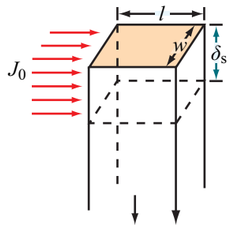
$$R_s = \frac{1}{\sigma\delta_s} = \sqrt{\frac{\pi f \mu}{\sigma}} \qquad (\Omega), \tag{7.92a}$$

$$L_s = \frac{1}{\omega\sigma\delta_s} = \frac{1}{2}\sqrt{\frac{\mu}{\pi f \sigma}} \qquad (\text{H}), \tag{7.92b}$$

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(a) Exponentially decaying $\tilde{J}_x(z)$



(b) Equivalent J_0 over skin depth δ_s

图 7-15 实心导体中电流密度 $\tilde{J}_x(z)$ 随 z 的指数衰减。流过 (a) 在 $z = 0$ 和 $z = \infty$ 之间延伸的宽度为 w 的截面的总电流相当于 (b) 流过深度为 δ_s 的截面的恒定电流密度 J_0。

导体中任意深度 z 处的电磁场由下式给出

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} E_0 e^{-\alpha z} e^{-j\beta z}, \tag{7.83a}$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z}. \tag{7.83b}$$

从 $\mathbf{J} = \sigma\mathbf{E}$ 开始，电流沿 x 方向流动，其密度为

$$\tilde{\mathbf{J}}(z) = \hat{\mathbf{x}}\, \tilde{J}_x(z), \tag{7.84}$$

和

$$\tilde{J}_x(z) = \sigma E_0 e^{-\alpha z} e^{-j\beta z} = J_0 e^{-\alpha z} e^{-j\beta z}, \tag{7.85}$$

其中 $J_0 = \sigma E_0$ 是表面电流密度的幅度。就趋肤深度 $\delta_s = 1/\alpha$ 而言，由方程定义： (7.72) 并利用良导体 $\alpha = \beta$ 如方程所示（7.77b），等式。(7.85) 可以写成

$$\tilde{J}_x(z) = J_0 e^{-(1+j)z/\delta_s} \qquad (\text{A/m}^2). \tag{7.86}$$

沿 y 方向流经宽度为 w 的矩形带并沿 z 方向在 0 到 ∞ 之间延伸的电流为

$$\tilde{I} = w \int_0^\infty \tilde{J}_x(z)\, dz = w \int_0^\infty J_0 e^{-(1+j)z/\delta_s}\, dz = \frac{J_0 w \delta_s}{(1+j)} \quad (\text{A}). \tag{7.87}$$

方程的分子。(7.87) 让人想起流过宽度为 w 、深度为 δ_s 的薄表面的均匀电流密度 J_0。因为 $\tilde{J}_x(z)$ 随深度 z 呈指数减小，只要 d 超过几个趋肤深度，则可以将有限厚度 d 的导体视为电等效于无限深度的导体。事实上，如果 $d = 3\delta_s$ [而不是等式积分中的 ∞ (7.87)]，使用等式右侧结果所产生的误差。(7.87)小于5%；如果$d = 5\delta_s$，则误差小于1%。

表面长度 l 上的电压（图 7-15(b)）由下式给出

$$\tilde{V} = E_0 l = \frac{J_0}{\sigma} l. \tag{7.88}$$

因此，宽度为 w、长度为 l、深度为 $d = \infty$ (或实际上为 $d > 5\delta_s$) 的板的阻抗为

$$Z = \frac{\tilde{V}}{\tilde{I}} = \frac{1+j}{\sigma\delta_s} \frac{l}{w} \qquad (\Omega)。 \tag{7.89}$$

通常将 Z 表示为

$$Z = Z_s \frac{l}{w}, \tag{7.90}$$

其中 Z_s 称为导体的内部或表面阻抗，定义为长度 $l = 1$ m 和宽度 $w = 1$ m 的阻抗 Z。因此，

$$Z_s = \frac{1+j}{\sigma\delta_s} \qquad (\Omega). \tag{7.91}$$

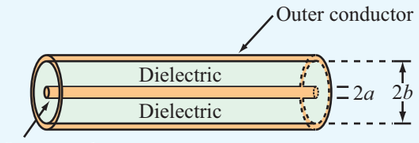
由于 Z_s 的电抗部分为正，因此 Z_s 可以定义为

$$Z_s = R_s + j\omega L_s$$

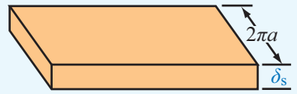
和

$$R_s = \frac{1}{\sigma\delta_s} = \sqrt{\frac{\pi f \mu}{\sigma}} \qquad (\Omega), \tag{7.92a}$$

$$L_s = \frac{1}{\omega\sigma\delta_s} = \frac{1}{2}\sqrt{\frac{\mu}{\pi f \sigma}} \qquad (\text{H}), \tag{7.92b}$$



(a) Coaxial cable



(b) Equivalent inner conductor

Figure 7-16 The inner conductor of the coaxial cable in (a) is represented in (b) by a planar conductor of width $2\pi a$ and depth δ_s, as if its skin has been cut along its length on the bottom side and then unfurled into a planar geometry.

where we used the relation $\delta_s = 1/\alpha \approx 1/\sqrt{\pi f \mu \sigma}$ given by Eq. (7.77a). In terms of the ***surface resistance*** R_s, the ***ac resistance*** of a slab of width w and length l is

$$R = R_s \frac{l}{w} = \frac{l}{\sigma \delta_s w} \quad (\Omega). \tag{7.93}$$

The expression for the ac resistance R is equivalent to the dc resistance of a plane conductor of length l and cross section $A = \delta_s w$.

The results obtained for the planar conductor can be extended to the coaxial cable shown in **Fig. 7-16(a)**. If the conductors are made of copper with $\sigma = 5.8 \times 10^7$ S/m, the skin depth at 1 MHz is $\delta_s = 1/\sqrt{\pi f \mu \sigma} = 0.066$ mm, and since δ_s varies as $1/\sqrt{f}$, it becomes smaller at higher frequencies. As long as the inner conductor's radius a is greater than $5\delta_s$, or 0.33 mm at 1 MHz, its "depth" may be regarded as infinite. A similar criterion applies to the thickness of the outer conductor. To compute the resistance of the inner conductor, note that the current is concentrated near its outer surface and approximately equivalent to a uniform current flowing through a thin layer of depth δ_s and circumference $2\pi a$. In other words, the inner conductor's resistance is nearly the same as that of a planar conductor of depth δ_s and width $w = 2\pi a$, as shown in **Fig. 7-16(b)**. The corresponding resistance per unit length is obtained by setting $w = 2\pi a$ and dividing by l in Eq. (7.93):

$$R'_1 = \frac{R}{l} = \frac{R_s}{2\pi a} \quad (\Omega/\text{m}). \tag{7.94}$$

Similarly, for the outer conductor, the current is concentrated within a thin layer of depth δ_s on the inside surface of the conductor adjacent to the insulating medium between the two conductors, which is where the EM fields exist. The resistance per unit length for the outer conductor with radius b is

$$R'_2 = \frac{R_s}{2\pi b} \quad (\Omega/\text{m}), \tag{7.95}$$

and the coaxial cable's total ac resistance per unit length is

$$R' = R'_1 + R'_2 = \frac{R_s}{2\pi}\left(\frac{1}{a} + \frac{1}{b}\right) \quad (\Omega/\text{m}). \tag{7.96}$$

This expression was used in Chapter 2 for characterizing the resistance per unit length of a coaxial transmission line.

Concept Question 7-6: How does β of a low-loss dielectric medium compare to that of a lossless medium?

Concept Question 7-7: In a good conductor, does the phase of **H** lead or lag that of **E** and by how much?

Concept Question 7-8: Attenuation means that a wave loses energy as it propagates in a lossy medium. What happens to the lost energy?

Concept Question 7-9: Is a conducting medium dispersive or dispersionless? Explain.

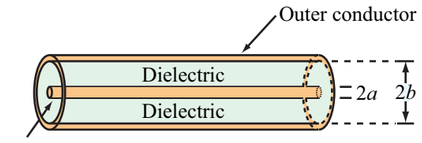
Concept Question 7-10: Compare the flow of current through a wire in the dc and ac cases. Compare the corresponding dc and ac resistances of the wire.

7-6 Electromagnetic Power Density

This section deals with the flow of power carried by an electromagnetic wave. For any wave with an electric field **E** and magnetic field **H**, the ***Poynting vector*** **S** is defined as

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\text{W/m}^2). \tag{7.97}$$

The unit of **S** is (V/m) × (A/m) = (W/m^2), and the direction of **S** is along the wave's direction of propagation. Thus, **S** represents the power per unit area (or ***power density***) carried by the wave. If the wave is incident upon an aperture of area A with outward surface unit vector $\hat{\mathbf{n}}$ as shown in **Fig. 7-17**,



(a) Coaxial cable



(b) Equivalent inner conductor

图 7-16 (a) 中同轴电缆的内导体在 (b) 中由宽度为 2π a 和深度为 δ_s 的平面导体表示，就好像其外皮已在底侧沿其长度被切割，然后展开成平面几何形状。

其中我们使用等式给出的关系 $\delta_s = 1/\alpha \approx 1/\sqrt{\pi f \mu \sigma}$ （7.77a）。就表面电阻 R_s 而言，宽度为 w、长度为 l 的平板的交流电阻为

$$R = R_s \frac{l}{w} = \frac{l}{\sigma \delta_s w} \quad (\Omega). \tag{7.93}$$

交流电阻 R 的表达式相当于长度为 l、横截面为 $A = \delta_s w$ 的平面导体的直流电阻。

针对平面导体获得的结果可以扩展到图7-16(a)所示的同轴电缆。如果导体由 $\sigma = 5.8 \times 10^7$ S/m 的铜制成，则 1 MHz 下的集肤深度为 $\delta_s = 1/\sqrt{\pi f \mu \sigma} = 0.066$ mm，并且由于 δ_s 变化为 $1/\sqrt{f}$，因此在较高频率下趋肤深度会变小。只要内导体的半径 a 大于 $5\delta_s$，即 1 MHz 时的 0.33 mm，其“深度”就可以被视为无限。类似的标准适用于外导体的厚度。要计算内导体的电阻，请注意电流集中在其外表面附近，大约相当于流过深度为 δ_s 且周长为 2π a 的薄层的均匀电流。换句话说，内导体的电阻与深度为δ_s、宽度为w= 2π a的平面导体的电阻几乎相同，如图7-16(b)所示。式(1)中设置w=2π a除以l即可得到相应的单位长度电阻。 （7.93）：

$$R'_1 = \frac{R}{l} = \frac{R_s}{2\pi a} \quad (\Omega/\text{m}). \tag{7.94}$$

类似地，对于外导体，电流集中在与两个导体之间的绝缘介质相邻的导体内表面上深度为 δ_s 的薄层内，这就是电磁场存在的地方。半径为 b 的外导体单位长度的电阻为

$$R'_2 = \frac{R_s}{2\pi b} \quad (\Omega/\text{m}), \tag{7.95}$$

同轴电缆单位长度的总交流电阻为

$$R' = R'_1 + R'_2 = \frac{R_s}{2\pi}\left(\frac{1}{a} + \frac{1}{b}\right) \quad (\Omega/\text{m}). \tag{7.96}$$

该表达式在第 2 章中用于表征同轴传输线单位长度的电阻。

Concept Question 7-6: How does β of a low-loss dielectric medium compare to that of a lossless medium?

Concept Question 7-7: In a good conductor, does the phase of **H** lead or lag that of **E** and by how much?

Concept Question 7-8: Attenuation means that a wave loses energy as it propagates in a lossy medium. What happens to the lost energy?

Concept Question 7-9: Is a conducting medium dispersive or dispersionless? Explain.

Concept Question 7-10: Compare the flow of current through a wire in the dc and ac cases. Compare the corresponding dc and ac resistances of the wire.

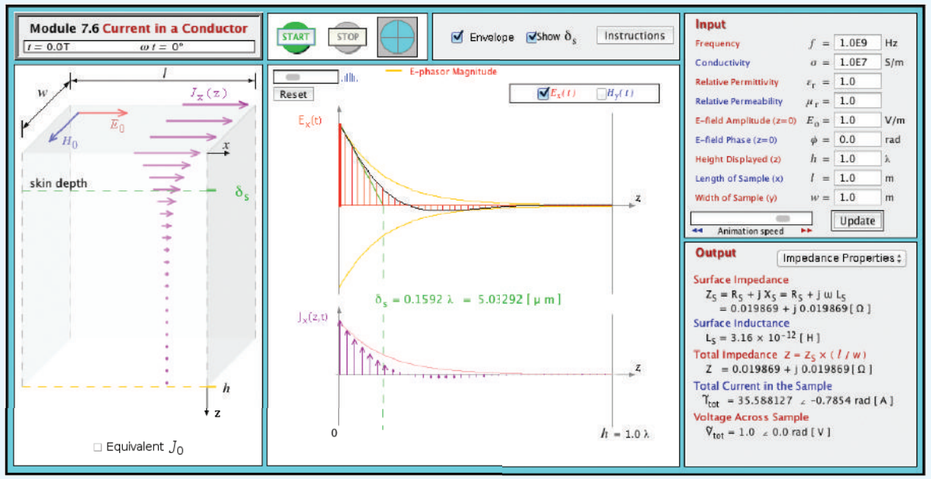
7-6 电磁功率密度

本节讨论电磁波携带的功率流。对于任何具有电场 E 和磁场 H 的波，坡印廷矢量 S 定义为

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\text{W/m}^2). \tag{7.97}$$

S的单位是(V/m)× (A/m) = (W/m^2)，S的方向沿着波的传播方向。因此，S代表波所携带的每单位面积的功率（或功率密度）。如果波以向外表面单位矢量n̂入射到区域A的孔径上，如图7-17所示，

Module 7.6 Current in a Conductor Module displays exponential decay of current density in a conductor.



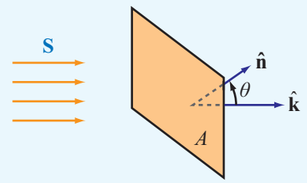


Figure 7-17 EM power flow through an aperture.

then the total power that flows through or is intercepted by the aperture is

$$P = \int_A \mathbf{S}\cdot\hat{\mathbf{n}}\,dA \qquad \text{(W)}. \tag{7.98}$$

For a uniform plane wave propagating in a direction $\hat{\mathbf{k}}$ that makes an angle θ with $\hat{\mathbf{n}}$, $P = SA\cos\theta$, where $S = |\mathbf{S}|$.

Except for the fact that the units of $\mathbf{S}$ are per unit area, Eq. (7.97) is the vector analogue of the scalar expression for the instantaneous power $P(z,t)$ flowing through a transmission line,

$$P(z,t) = v(z,t)\,i(z,t), \tag{7.99}$$

where $v(z,t)$ and $i(z,t)$ are the instantaneous voltage and current on the line.

Since both $\mathbf{E}$ and $\mathbf{H}$ are functions of time, so is the Poynting vector $\mathbf{S}$. In practice, however, the quantity of greater interest is the ***average power density*** of the wave, $\mathbf{S}_{av}$, which is the time-average value of $\mathbf{S}$:

$$\mathbf{S}_{av} = \tfrac{1}{2}\,\mathfrak{Re}\left[\widetilde{\mathbf{E}}\times\widetilde{\mathbf{H}}^*\right] \qquad (\text{W/m}^2). \tag{7.100}$$

This expression may be regarded as the electromagnetic equivalent of Eq. (2.107) for the time-average power carried by a transmission line, namely

$$P_{av}(z) = \tfrac{1}{2}\,\mathfrak{Re}\left[\widetilde{V}(z)\,\widetilde{I}^*(z)\right], \tag{7.101}$$

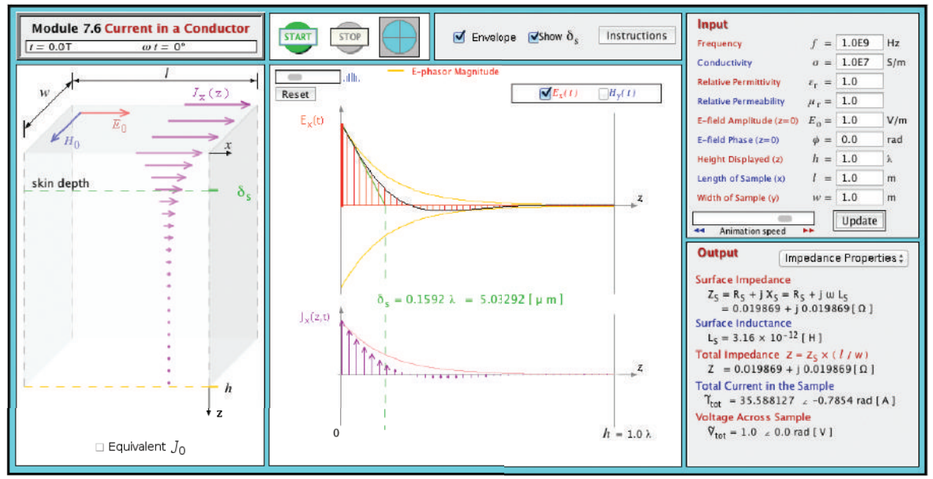
where $\widetilde{V}(z)$ and $\widetilde{I}(z)$ are the phasors corresponding to $v(z,t)$ and $i(z,t)$, respectively.

7-6.1 Plane Wave in a Lossless Medium

Recall that the general expression for the electric field of a uniform plane wave with arbitrary polarization traveling in the $+z$ direction is

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\,\widetilde{E}_x(z) + \hat{\mathbf{y}}\,\widetilde{E}_y(z) = (\hat{\mathbf{x}}E_{x0} + \hat{\mathbf{y}}E_{y0})e^{-jkz}, \tag{7.102}$$

模块 7.6 导体中的电流 模块显示导体中电流密度的指数衰减。



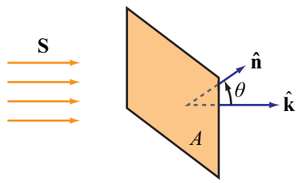


图 7-17 通过孔径的电磁功率流。

那么流经孔径或被孔径拦截的总功率为

$$P = \int_A \mathbf{S}\cdot\hat{\mathbf{n}}\,dA \qquad \text{(W)}. \tag{7.98}$$

对于沿方向 k̂ 传播且与 n̂ 形成角度 θ 的均匀平面波，P = SA cosθ，其中 S = |S|。

除了 S 的单位是每单位面积这一事实外，方程：(7.97) 是流经传输线的瞬时功率 P(z,t) 标量表达式的矢量模拟，

$$P(z,t) = v(z,t)\,i(z,t), \tag{7.99}$$

其中 v(z,t) 和 i(z,t) 是线路上的瞬时电压和电流。

由于 E 和 H 都是时间的函数，因此坡印廷向量 S 也是时间的函数。然而，在实践中，更令人感兴趣的量是波的平均功率密度 S_{av}，它是 S 的时间平均值：

$$\mathbf{S}_{av} = \tfrac{1}{2}\,\mathfrak{Re}\left[\widetilde{\mathbf{E}}\times\widetilde{\mathbf{H}}^*\right] \qquad (\text{W/m}^2). \tag{7.100}$$

这个表达式可以被认为是方程的电磁等价。(2.107) 为传输线承载的时间平均功率，即

$$P_{av}(z) = \tfrac{1}{2}\,\mathfrak{Re}\left[\widetilde{V}(z)\,\widetilde{I}^*(z)\right], \tag{7.101}$$

其中 ~V (z) 和 ~I(z) 分别是对应于 v(z,t) 和 i(z,t) 的相量。

7-6.1 无损介质中的平面波

回想一下，沿 +z 方向传播的具有任意偏振的均匀平面波的电场的一般表达式为

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\,\widetilde{E}_x(z) + \hat{\mathbf{y}}\,\widetilde{E}_y(z) = (\hat{\mathbf{x}}E_{x0} + \hat{\mathbf{y}}E_{y0})e^{-jkz}, \tag{7.102}$$

where, in the general case, E_{x0} and E_{y0} may be complex quantities. The magnitude of $\widetilde{\mathbf{E}}$ is

$$|\widetilde{\mathbf{E}}| = (\widetilde{\mathbf{E}} \cdot \widetilde{\mathbf{E}}^*)^{1/2} = [|E_{x0}|^2 + |E_{y0}|^2]^{1/2}. \tag{7.103}$$

The phasor magnetic field associated with $\widetilde{\mathbf{E}}$ is obtained by applying Eq. (7.39a):

$$\widetilde{\mathbf{H}}(z) = (\hat{\mathbf{x}}\widetilde{H}_x + \hat{\mathbf{y}}\widetilde{H}_y)e^{-jkz} = \frac{1}{\eta}\hat{\mathbf{z}} \times \widetilde{\mathbf{E}} = \frac{1}{\eta}(-\hat{\mathbf{x}}E_{y0} + \hat{\mathbf{y}}E_{x0})e^{-jkz}. \tag{7.104}$$

The wave can be considered as the sum of two waves: one comprising fields $(\widetilde{E}_x, \widetilde{H}_y)$ and another comprising fields $(\widetilde{E}_y, \widetilde{H}_x)$. Use of Eqs. (7.102) and (7.104) in Eq. (7.100) leads to

$$\mathbf{S}_{\text{av}} = \hat{\mathbf{z}}\frac{1}{2\eta}(|E_{x0}|^2 + |E_{y0}|^2) = \hat{\mathbf{z}}\frac{|\widetilde{\mathbf{E}}|^2}{2\eta} \quad (\text{W/m}^2), \tag{7.105}$$

(lossless medium)

which states that power flows in the z direction with average power density equal to the sum of the average power densities of the $(\widetilde{E}_x, \widetilde{H}_y)$ and $(\widetilde{E}_y, \widetilde{H}_x)$ waves. Note that, because $\mathbf{S}_{\text{av}}$ depends only on η and $|\widetilde{\mathbf{E}}|$, waves characterized by different polarizations carry the same amount of average power as long as their electric fields have the same magnitudes.

Example 7-5: Solar Power

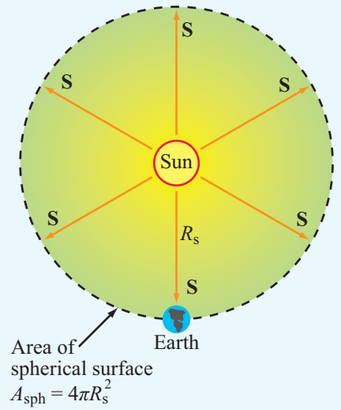
If solar illumination is characterized by a power density of 1 kW/m^2 on the Earth's surface, find (a) the total power radiated by the sun, (b) the total power intercepted by the Earth, and (c) the electric field of the power density incident upon the Earth's surface, assuming that all the solar illumination is at a single frequency. The radius of the Earth's orbit around the sun, R_s, is approximately 1.5×10^8 km, and the Earth's mean radius R_e is 6,380 km.

Solution: **(a)** Assuming that the sun radiates isotropically (equally in all directions), the total power it radiates is $S_{\text{av}} A_{\text{sph}}$, where A_{sph} is the area of a spherical shell of radius R_s (**Fig. 7-18(a)**). Thus,

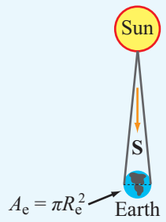
$$P_{\text{sun}} = S_{\text{av}}(4\pi R_s^2) = 1 \times 10^3 \times 4\pi \times (1.5 \times 10^{11})^2 = 2.8 \times 10^{26} \text{ W}.$$

(b) With reference to **Fig. 7-18(b)**, the power intercepted by the Earth's cross section $A_e = \pi R_e^2$ is

$$P_{\text{int}} = S_{\text{av}}(\pi R_e^2) = 1 \times 10^3 \times \pi \times (6.38 \times 10^6)^2 = 1.28 \times 10^{17} \text{ W}.$$



(a) Radiated solar power



(b) Earth intercepted power

Figure 7-18 Solar radiation intercepted by (a) a spherical surface of radius R_s and (b) the Earth's surface (Example 7-5).

(c) The power density S_{av} is related to the magnitude of the electric field $|\widetilde{E}| = E_0$ by

$$S_{\text{av}} = \frac{E_0^2}{2\eta_0},$$

where $\eta_0 = 377\ (\Omega)$ for air. Hence,

$$E_0 = \sqrt{2\eta_0 S_{\text{av}}} = \sqrt{2 \times 377 \times 10^3} = 870 \quad (\text{V/m}).$$

7-6.2 Plane Wave in a Lossy Medium

The expressions given by Eqs. (7.68) and (7.69) characterize the electric and magnetic fields of an x-polarized plane wave

其中，在一般情况下，E_{x0}和E_{y0}可以是复数。 $\widetilde{\mathbf{E}}$ 的大小为

$$|\widetilde{\mathbf{E}}| = (\widetilde{\mathbf{E}} \cdot \widetilde{\mathbf{E}}^*)^{1/2} = [|E_{x0}|^2 + |E_{y0}|^2]^{1/2}. \tag{7.103}$$

与 $\widetilde{\mathbf{E}}$ 相关的相量磁场可通过应用等式获得。 (7.39a)：

$$\widetilde{\mathbf{H}}(z) = (\hat{\mathbf{x}}\widetilde{H}_x + \hat{\mathbf{y}}\widetilde{H}_y)e^{-jkz} = \frac{1}{\eta}\hat{\mathbf{z}} \times \widetilde{\mathbf{E}} = \frac{1}{\eta}(-\hat{\mathbf{x}}E_{y0} + \hat{\mathbf{y}}E_{x0})e^{-jkz}. \tag{7.104}$$

该波可以被视为两个波的总和：一个波包含场（$\widetilde{E}_x$，$\widetilde{H}_y$），另一个波包含场（$\widetilde{E}_y$，$\widetilde{H}_x$）。使用等式。式（7.102）和（7.104）(7.100) 导致

$$\mathbf{S}_{\text{av}} = \hat{\mathbf{z}}\frac{1}{2\eta}(|E_{x0}|^2 + |E_{y0}|^2) = \hat{\mathbf{z}}\frac{|\widetilde{\mathbf{E}}|^2}{2\eta} \quad (\text{W/m}^2), \tag{7.105}$$

(lossless medium)

其中指出功率沿 z 方向流动，平均功率密度等于 ($\widetilde{E}_x$, $\widetilde{H}_y$) 和 ($\widetilde{E}_y$, $\widetilde{H}_x$) 波的平均功率密度之和。请注意，由于 S_{av} 仅取决于 η 和 $|\widetilde{E}|$，因此只要电场强度相同，具有不同偏振特征的波就会携带相同的平均功率。

例7-5：太阳能

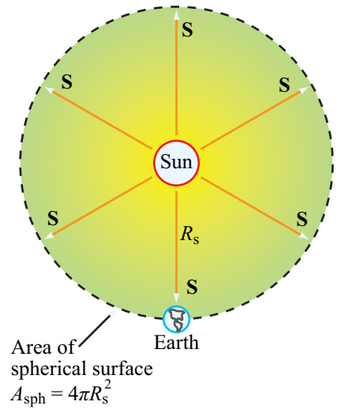
如果太阳光照的特征是地球表面的功率密度为 1 kW/m^2，则假设所有太阳光照都处于单一频率，则求 (a) 太阳辐射的总功率，(b) 地球截获的总功率，以及 (c) 入射到地球表面的功率密度的电场。地球绕太阳轨道的半径 R_s 约为 1.5×10^8 公里，地球平均半径 R_e 为 6,380 公里。

解：(a) 假设太阳各向同性（向各个方向均匀）辐射，则其辐射的总功率为 $S_{av} A_{sph}$，其中 A_{sph} 是半径为 R_s (的球壳面积。 7-18（a））。因此，

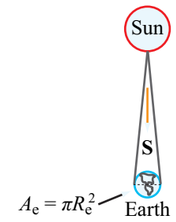
$$P_{\text{sun}} = S_{\text{av}}(4\pi R_s^2) = 1 \times 10^3 \times 4\pi \times (1.5 \times 10^{11})^2 = 2.8 \times 10^{26} \text{ W}.$$

(b) 参考图7-18(b)，地球横截面截获的功率$A_e = \pi R_e^2$为

$$P_{\text{int}} = S_{\text{av}}(\pi R_e^2) = 1 \times 10^3 \times \pi \times (6.38 \times 10^6)^2 = 1.28 \times 10^{17} \text{ W}.$$



(a) Radiated solar power



(b) Earth intercepted power

图 7-18 (a) 半径为 R_s 的球面和 (b) 地球表面截获的太阳辐射（例 7-5）。

(c) 功率密度 S_{av} 与电场 $|\widetilde{E}| = E_0$ 的大小相关：

$$S_{\text{av}} = \frac{E_0^2}{2\eta_0},$$

其中 $\eta_0 = 377\ (\Omega)$ 代表空气。因此，

$$E_0 = \sqrt{2\eta_0 S_{\text{av}}} = \sqrt{2 \times 377 \times 10^3} = 870 \quad (\text{V/m}).$$

7-6.2 有损介质中的平面波

等式给出的表达式。 (7.68) 和 (7.69) 描述了 x 偏振平面波的电场和磁场

propagating along the z direction in a lossy medium with propagation constant $\gamma = \alpha + j\beta$. By extending these expressions to the more general case of a wave with components along both x and y, we have

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\,\widetilde{E}_x(z) + \hat{\mathbf{y}}\,\widetilde{E}_y(z) = (\hat{\mathbf{x}}E_{x0} + \hat{\mathbf{y}}E_{y0})e^{-\alpha z}e^{-j\beta z}, \tag{7.106a}$$

$$\widetilde{\mathbf{H}}(z) = \frac{1}{\eta_c}(-\hat{\mathbf{x}}E_{y0} + \hat{\mathbf{y}}E_{x0})e^{-\alpha z}e^{-j\beta z}, \tag{7.106b}$$

where η_c is the intrinsic impedance of the lossy medium. Application of Eq. (7.100) gives

$$\mathbf{S}_{av}(z) = \frac{1}{2}\,\mathfrak{Re}\left[\widetilde{\mathbf{E}} \times \widetilde{\mathbf{H}}^*\right] = \frac{\hat{\mathbf{z}}(|E_{x0}|^2 + |E_{y0}|^2)}{2}e^{-2\alpha z}\,\mathfrak{Re}\left(\frac{1}{\eta_c^*}\right). \tag{7.107}$$

By expressing η_c in polar form as

$$\eta_c = |\eta_c|e^{j\theta_\eta}, \tag{7.108}$$

Eq. (7.107) can be rewritten as

$$\mathbf{S}_{av}(z) = \hat{\mathbf{z}}\,\frac{|\widetilde{E}(0)|^2}{2|\eta_c|}\,e^{-2\alpha z}\cos\theta_\eta \ \ (\mathrm{W/m^2}), \tag{7.109}$$

(lossy medium)

where $|\widetilde{E}(0)|^2 = [|E_{x0}|^2 + |E_{y0}|^2]^{1/2}$ is the magnitude of $\widetilde{\mathbf{E}}(z)$ at $z = 0$.

▶ Whereas the fields $\widetilde{\mathbf{E}}(z)$ and $\widetilde{\mathbf{H}}(z)$ decay with z as $e^{-\alpha z}$, the power density $\mathbf{S}_{av}$ decreases as $e^{-2\alpha z}$. ◀

When a wave propagates through a distance $z = \delta_s = 1/\alpha$, the magnitudes of its electric and magnetic fields decrease to $e^{-1} \approx 37\%$ of their initial values, and its average power density decreases to $e^{-2} \approx 14\%$ of its initial value.

7-6.3 Decibel Scale for Power Ratios

The unit for power P is watts (W). In many engineering problems, the quantity of interest is the ratio of two power levels, P_1 and P_2, such as the incident and reflected powers on a transmission line, and often the ratio P_1/P_2 may vary over several orders of magnitude. The decibel (dB) scale is logarithmic, thereby providing a convenient representation of the power ratio, particularly when numerical values of P_1/P_2 are plotted against some variable of interest. If

$$G = \frac{P_1}{P_2}, \tag{7.110}$$

then

$$G\ [\mathrm{dB}] = 10\log G = 10\log\left(\frac{P_1}{P_2}\right) \quad (\mathrm{dB}). \tag{7.111}$$

Table 7-2 Power ratios in natural numbers and in decibels.

G	G [dB]
10^x	$10x$ dB
4	6 dB
2	3 dB
1	0 dB
0.5	−3 dB
0.25	−6 dB
0.1	−10 dB
10^{-3}	−30 dB

Table 7-2 provides a comparison between values of G and the corresponding values of G [dB]. Even though decibels are defined for power ratios, they can sometimes be used to represent other quantities. For example, if $P_1 = V_1^2/R$ is the power dissipated in a resistor R with voltage V_1 across it at time t_1, and $P_2 = V_2^2/R$ is the power dissipated in the same resistor at time t_2, then

$$\begin{aligned} G\ [\mathrm{dB}] &= 10\log\left(\frac{P_1}{P_2}\right) \\ &= 10\log\left(\frac{V_1^2/R}{V_2^2/R}\right) \\ &= 20\log\left(\frac{V_1}{V_2}\right) \\ &= 20\log(g) \\ &= g\ [\mathrm{dB}], \end{aligned} \tag{7.112}$$

where $g = V_1/V_2$ is the voltage ratio. *Note that for voltage (or current) ratios the scale factor is 20 rather than 10,* which results in G [dB] $= g$ [dB].

The ***attenuation rate***, representing the rate of decrease of the magnitude of $\mathbf{S}_{av}(z)$ as a function of propagation distance,

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在有损介质中沿 z 方向传播，传播常数为 $\gamma = \alpha + j\beta$。通过将这些表达式扩展到具有沿 x 和 y 分量的波的更一般情况，我们有

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\,\widetilde{E}_x(z) + \hat{\mathbf{y}}\,\widetilde{E}_y(z) = (\hat{\mathbf{x}}E_{x0} + \hat{\mathbf{y}}E_{y0})e^{-\alpha z}e^{-j\beta z}, \tag{7.106a}$$

$$\widetilde{\mathbf{H}}(z) = \frac{1}{\eta_c}(-\hat{\mathbf{x}}E_{y0} + \hat{\mathbf{y}}E_{x0})e^{-\alpha z}e^{-j\beta z}, \tag{7.106b}$$

其中 η_c 是有损介质的固有阻抗。方程的应用(7.100) 给出

$$\mathbf{S}_{av}(z) = \frac{1}{2}\,\mathfrak{Re}\left[\widetilde{\mathbf{E}} \times \widetilde{\mathbf{H}}^*\right] = \frac{\hat{\mathbf{z}}(|E_{x0}|^2 + |E_{y0}|^2)}{2}e^{-2\alpha z}\,\mathfrak{Re}\left(\frac{1}{\eta_c^*}\right). \tag{7.107}$$

将 η_c 以极坐标形式表示为

$$\eta_c = |\eta_c|e^{j\theta_\eta}, \tag{7.108}$$

等式。 (7.107) 可以重写为

$$\mathbf{S}_{av}(z) = \hat{\mathbf{z}}\,\frac{|\widetilde{E}(0)|^2}{2|\eta_c|}\,e^{-2\alpha z}\cos\theta_\eta \ \ (\mathrm{W/m^2}), \tag{7.109}$$

(lossy medium)

其中 $|\widetilde{E}(0)|^2 = [|E_{x0}|^2 + |E_{y0}|^2]^{1/2}$ 是 $\widetilde{E}(z)$ 在 $z = 0$ 处的幅度。

▶ 当 z 为 $e^{-\alpha z}$ 时，场 $\widetilde{E}(z)$ 和 $\widetilde{H}(z)$ 衰减，而功率密度 S_{av} 则随着 $e^{-2\alpha z}$ 降低。 ◀

当波传播距离 $z = \delta_s = 1/\alpha$ 时，其电场和磁场的强度减小至初始值的 $e^{-1} \approx 37\%$，平均功率密度减小至初始值的 $e^{-2} \approx 14\%$。

7-6.3 功率比的分贝标度

功率P的单位是瓦（W）。在许多工程问题中，感兴趣的量是两个功率电平 P_1 和 P_2 的比率，例如传输线上的入射功率和反射功率，并且通常比率 P_1/P_2 可能会变化几个数量级。分贝 (dB) 刻度是对数的，从而可以方便地表示

表 7-2 以自然数和分贝为单位的功率比。

G	G [dB]
10^x	$10x$ dB
4	6 dB
2	3 dB
1	0 dB
0.5	−3 dB
0.25	−6 dB
0.1	−10 dB
10^{-3}	−30 dB

功率比，特别是当 P_1/P_2 的数值针对某些感兴趣的变量绘制时。如果

$$G = \frac{P_1}{P_2}, \tag{7.110}$$

然后

$$G\ [\mathrm{dB}] = 10\log G = 10\log\left(\frac{P_1}{P_2}\right) \quad (\mathrm{dB}). \tag{7.111}$$

表 7-2 提供了 G 值和相应的 G [dB] 值之间的比较。尽管分贝是针对功率比定义的，但有时它们也可以用来表示其他量。例如，如果 $P_1 = V_1^2/R$ 是在时间 t_1 时电阻器 R 上的电压为 V_1 时消耗的功率，而 $P_2 = V_2^2/R$ 是在时间 t_2 时同一电阻器中消耗的功率，则

$$\begin{aligned} G\ [\mathrm{dB}] &= 10\log\left(\frac{P_1}{P_2}\right) \\ &= 10\log\left(\frac{V_1^2/R}{V_2^2/R}\right) \\ &= 20\log\left(\frac{V_1}{V_2}\right) \\ &= 20\log(g) \\ &= g\ [\mathrm{dB}], \end{aligned} \tag{7.112}$$

其中 $g = V_1/V_2$ 是电压比。请注意，对于电压（或电流）比，比例因子为 20 而不是 10，结果为 G [dB] = g [dB]。

衰减率，表示 $S_{av}(z)$ 幅度随传播距离变化的减小率，